



Probing Layer-wise Alignment Depth in Multimodal Transformers

Final Presentation

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Overview

ML Models combine different modalities

- Utilize information from multiple modalities like text and vision
- Theoretically hard to combine different formats
- Transformers implicitly learn to bring modalities together
- **Goal of this thesis:** Understand where alignment happens



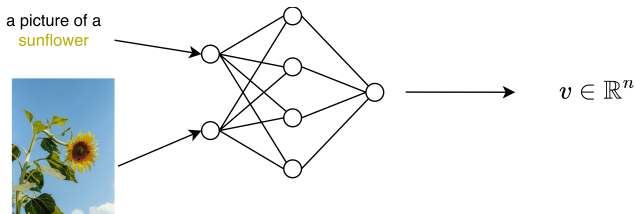
[3].

Vegan Apple Pies:
Recipe for Apple Pie

Representation Learning

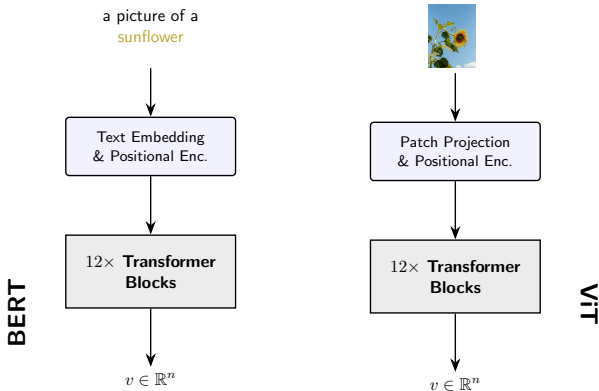
Goal: Generate useful vector representations from multiple modalities

- Vector used for downstream tasks (vector DB, classification, regression, ...)
- Vectors have high dimensionality ($n = 768$)
- Encoder-only Transformer



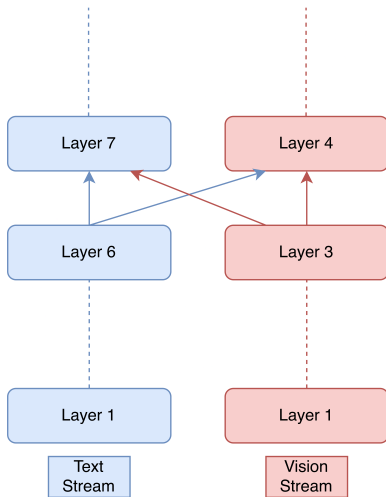
[1]

Text and Vision Encoder



Dual-Stream Architecture (ViLBERT)

- Two streams: vision (ViT) and text (BERT)
- 12 transformer blocks/stream
- Cross-attention bridges the two streams
- **Key question:** where should cross-attention be placed?
 - Early Fusion: layers 4-6
 - Middle Fusion: layers 7-9
 - Late Fusion: layers 10-12



Methodology

Analysis

- 8 cross-attention configurations pretrained + finetuned
- Measure alignment metrics (CKA, SVCCA, mkNN, OP) at every layer

Three datasets with varying modality redundancy:

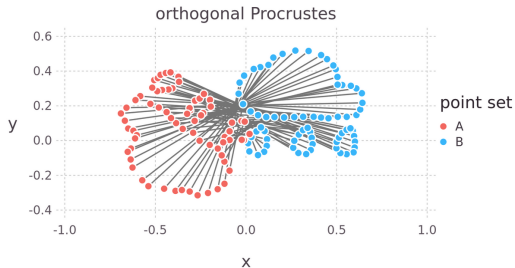
- *UPMC-Food101*: recipes + food images, high redundancy (easy)
- *MM-IMDb*: movie plots + posters, moderate redundancy
- *Hateful Memes*: text + image, often contradictory (hard)

Representational Alignment

Quantitative Analysis of Alignment

- (Dis)similarity metric to compare two representation spaces
- Used four metrics (Orthogonal Procrustes, SVCCA, mkNN, CKA)
- Structural and geometric measures

Example: Orthogonal Procrustes: $\min_{\mathbf{Q}} \|\mathbf{A}\mathbf{Q} - \mathbf{B}\|_F$



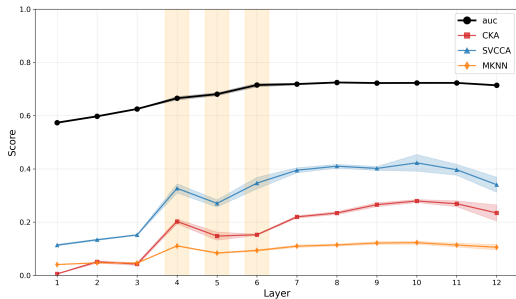
Source: [2]

Late Fusion Outperforms Across All Tasks

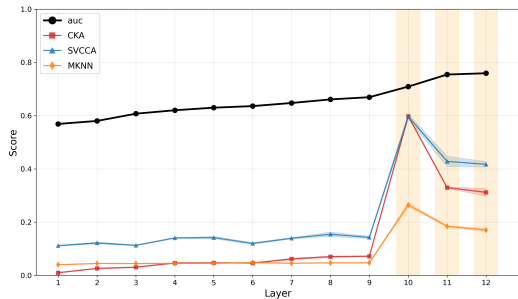
Model	HM (AUROC)	MM-IMDb (F1)	Food101 (Acc.)
No Cross-Attn	0.654	0.483	0.887
Early Fusion	0.723	0.512	0.893
Middle Fusion	0.750	0.533	0.918
Late Fusion	0.764	0.545	0.928
Full Cross-Attn	0.707	0.528	0.909

Table: n=3 seeds per configuration

Cross-Attention Increases Alignment



(a) Early Fusion: moderate representational alignment



(b) Late Fusion: sharp alignment spike at layers 10–12

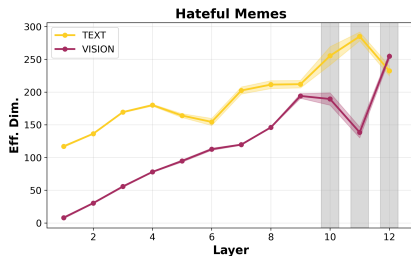
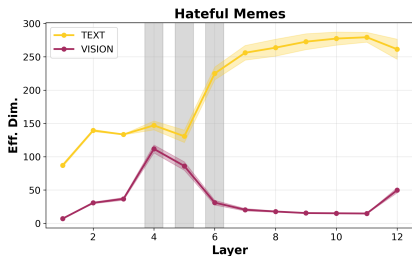
Later Fusion $\Rightarrow$ Higher alignment

Why Does Late Fusion Win?

Experiment: Effective Dimensionality (PCA keeping 95% of variance)

Key Finding: Early Fusion causes **dimensionality collapse** in vision stream

- **Dimensionality Drop:** Vision variance collapses from ~ 150 to ~ 20 principal components
- Late Fusion *preserves representations* until both modalities are mature



Collapse Hypothesis

Discrepancies between text and image tokenizations

- Text tokens have high semantic information, vision patches are noisy
- Image patches are noisy and high entropy
- ViT needs more layers to extract high-level features
- Early Fusion forces alignment **before vision is ready**



(a) Visual Embeddings (Patches)

This is an example for tokenization



(b) Text Embeddings (Tokens)

Summary

- **Late Fusion consistently outperforms** across all tasks (2–5% gain)
- Cross-attention **sharply increases** alignment: later placement, stronger effect
- Early Fusion causes **permanent dimensionality collapse** in the vision stream
- Alignment–performance correlation is **task-dependent**: strong for redundant modalities, weak for complex tasks like Hateful Memes



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Multimodal machine learning lecture.

<https://piazza.com/cmu/fall2023/11777/resources>, 2023.

[Accessed 09-06-2025].



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The orthogonal procrustes problem, October 2018.

Blog post.



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Recipe recognition with large multimodal food dataset.

In *2015 IEEE International Conference on Multimedia & Expo Workshops (ICMEW)*, pages 1–6, June 2015.